Final Review Problems

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Gronwall Inequality

Problem

Let $y : [0, T] \to \mathbb{R}$ satisfy

$$y'(t) \leq 3y(t) + 2, \qquad y(0) = y_0.$$

1. Use an integrating factor to derive an explicit upper bound of the form

$$y(t) \leq C_1 e^{3t} + C_2$$

for all $t \in [0, T]$, with C_1, C_2 in terms of y_0.

2. Re-derive the same bound using the integral inequality

$$y(t) \leq y_0 + \int_0^t \big(3y(s) + 2\big)\, ds$$

together with Grönwall's inequality.

Comparison Principle

Problem

Consider

$$y'(t) \le 2t\,y(t) + t^2, \qquad y(0) = 0,$$

and the linear ODE

$$x'(t) = 2t\,x(t) + t^2, \qquad x(0) = 0.$$

1. Solve for $x(t)$ explicitly.
2. Use the comparison principle to show $y(t) \le x(t)$ for all $t \ge 0$.
3. Check that this upper bound is the same as what you obtain by directly applying Grönwall's inequality to the inequality for y.

Finite-Time Blowup

Problem

Consider the ODE

$$y' = y^3, \qquad y(0) = 1.$$

1. Solve explicitly and show that the solution blows up in finite time.
2. For the sequence $t_n = T - \frac{1}{n}$ where T is the blow-up time, compute $y(t_n)$.
3. Around each point $(t_n, y(t_n))$ apply the existence–uniqueness theorem in a rectangle and show that the guaranteed forward time of existence h_n satisfies $t_n + h_n < T$ for all n, with $h_n \to 0$ as $n \to \infty$.
4. Explain how this illustrates that the only obstruction to extending the solution beyond T is the unboundedness of $y(t)$.

Problem

Determine whether solutions are global for any initial condition; justify your answer in each case.

1. $y' = \dfrac{y^2}{1+t^2}$.
2. $y' = \dfrac{y}{1+y^2}$.

Linear Stability

Problem

Consider

$$\begin{cases} x' = x(1-y), \\ y' = y(x-2). \end{cases}$$

1. Find all equilibria.
2. Compute the Jacobian at each equilibrium and find its eigenvalues.
3. Classify each equilibrium (saddle, node, spiral, center) for the linearized system.
4. Use the nonlinear linearization theorem (for hyperbolic equilibria) to state the nonlinear stability type of each equilibrium in the full system. (Where linearization is not conclusive, say so and explain why.)

Lyapunov Functions

Problem

Consider

$$\begin{cases} x' = -x + 2y, \\ y' = -3x - y. \end{cases}$$

1. Show directly (by eigenvalues) that the origin is asymptotically stable.
2. Look for a Lyapunov function of the form

   $$V(x, y) = ax^2 + bxy + cy^2$$

   with $a > 0$, $c > 0$ and $4ac > b^2$.

   (i) Compute $\dot{V}$ along solutions and write it as a quadratic form in x, y.

   (ii) Choose a, b, c so that $\dot{V}$ is negative definite.
3. Conclude again that the origin is asymptotically stable by Lyapunov's direct method.

Limit Sets

Problem

Consider the planar system in polar coordinates:

$$r' = r(1 - r^2), \qquad \theta' = 2.$$

1. Find all equilibria of the scalar ODE for $r(t)$ and classify their stability.
2. Describe the corresponding periodic orbits in the plane (circles of what radii?).
3. For initial data with $r_0 > 0$, describe the ω-limit set of the trajectory in $\mathbb{R}^2$.
4. For $0 < r_0 < 1$, describe the α-limit set; for $r_0 > 1$, describe the α-limit set.